AI-Strike AI Pentesting Tool

Project Proposal

1. Executive Summary

AI-Assisted Pentesting Tool is a web-based penetration testing application that automates security assessment of a web application and uses AI as a co-pilot to interpret the results. The tool runs four automated scan modules: reconnaissance, SQL injection, cross-site scripting (XSS), and misconfiguration testing, against a target application under written authorization. Findings from each module are written into a shared JSON format, which an AI layer then prioritizes: assigning severity, explaining each finding in plain English, suggesting remediation fixes and follow-up payloads to help verify or deepen each finding, and generating a report summary Results are presented through a web dashboard with a final exportable report. The workflow is: recon discovers the attack surface, the three testing modules scan it in parallel, the AI layer analyzes and prioritizes the findings, and the dashboard displays and exports the results.

2. Technologies

Category	Tools / Languages
Core language	Python 3.12
Scanning / HTTP	requests, httpx, BeautifulSoup4
SQL injection testing	sqlmap (invoked via subprocess)
Backend API	FastAPI, Uvicorn
Database	SQLite
Frontend / Dashboard	HTML/CSS/JavaScript
AI layer	Ollama (local LLM) — Qwen2.5 7B-Instruct, with Qwen2.5-Coder 3B as a lightweight fallback.
Version control	Git, GitHub
Environment	Python venv, python-dotenv for secrets

3. Timeline

Project start date: 10 October. Final delivery date: 18 November.

Date	Milestone / Deliverables
10/10	● Project kickoff. ● Findings JSON schema finalized and agreed by all modules. ● Written pentest authorization obtained from the target application's owner. ● GitHub repository created; all team members set up per SETUP.md (Python, sqlmap, Git, dependencies installed and verified).
17/10	● Recon module functional: fingerprinting, endpoint crawling, and parameter extraction working against the target. ● FastAPI backend skeleton and dashboard shell in place, running against mock findings data.
24/10	● SQLi module functional (sqlmap integration and output parsing complete) and tested against recon's real output. ● Misconfig module functional (header checks, exposed-path checks, CSRF token and cookie-flag checks) and tested against recon's real output.
4/11	● XSS module functional: payload set built, reflection-detection logic working and tuned against false positives, tested against recon's real output. ● AI prompt templates drafted and tested against mock findings for triage, severity assignment, fix suggestions, and payload suggestions.
11/11	● Full integration: all four scan modules feeding real results into the API and dashboard. ● AI layer wired to real findings for prioritization, explanations, fix suggestions, and payload suggestions. ● First end-to-end scan executed against the live target application.
14/11	● Bugs and false positives from the first full run fixed and tuned. ● Report export feature (AI-generated summary plus findings list) implemented. ● Dashboard UI polished.
18/11	● Final full scan run performed to collect clean, verified findings. ● Written report finalized. ● Final delivery: completed tool, finalized report, and demo-ready presentation.

4. Team Roles

Name	Module	Responsibilities
Mariam Mohamed Aly	AI Layer	Designed and implemented the AI co-pilot layer: prompt engineering for evaluating findings, assigning severity, suggesting fixes and payloads, generating plain-English explanations, and producing the final report summary from the consolidated findings JSON.
Kenzy Hany Mohamed	API	Built the FastAPI backend connecting the scan modules, the AI layer, and the dashboard: scan trigger and status endpoints, findings retrieval, and report export.

Mohamed Ashraf Kasem	SQLi	Built the SQL injection testing module: integrated sqlmap via subprocess calls against recon-discovered parameters and parsed its output into the shared findings format.
Menna Ayman Nasr	XSS	Built the cross-site scripting testing module: developed the payload set and reflection-detection logic to identify unescaped payload reflection across discovered input points.
Mohsen Amr Mohsen	Misconfig	Built the misconfiguration testing module: security header checks, exposed file/path checks, CSRF token verification, and cookie attribute checks.
Abdelrahman Ahmed	Recon	Built the reconnaissance module: target fingerprinting, endpoint crawling, and parameter extraction, forming the foundation that all other scan modules rely on.